

Coordinated of multi-point and bi-casting joint soft handover scheme for high-speed rail

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Abstract: The high mobility of high-speed train brings great challenges to the current wireless communication, especially to the handover. How to reduce the interrupt time and enhance the handover performance is becoming a serious problem for high-speed rail (HSR). In this study, a coordinated of multi-point (CoMP) and bi-casting joint soft handover scheme for HSR is proposed, which is based on long-term evolution network architecture, and without any change to the existing network and the train. In the scheme, CoMP is used on air interface and bi-casting is used on S1 interface. With the binding triggering of CoMP and bi-casting, only one suit of signalling is needed and the system complexity is much simplified. Meanwhile, as much handover related information is stored in corresponding entity during the joining process of CoMP and bi-casting, the handover process is much simplified and converted into update of serving eNodeB. Compared with traditional handover scheme, it can be seen from theoretical analysis and simulation results that the handover delay is reduced remarkably and the handover performance is improved greatly.

1 Introduction

The moving speed of high-speed trains keeps increasing from 200 to 350 km/h, even up to 500 km/h in the near future. The rising speed provides much convenience to passengers, but it also brings great challenges to the current wireless communication. Firstly, the Doppler shift and fast fading channel characteristics caused by high-speed lead to a great threat to the high-speed data transmission. Especially for time division duplex (TDD) system, the uplink frequency offset is twice of the same case in downlink, which may result in difficulty in the decoding of received signals. Meanwhile, in handover region, the train will receive signals from two adjacent base stations, which are of the same energy but with a jumping from positive to negative in frequency offset. Secondly, the increase in speed will result in frequent handover, which will spend a higher cost on mobility management, especially fast handover control. Finally, as the passengers on board are geographically located relatively concentrated, group handover will occur when the train goes through the cell edge, which is also a problem rarely faced, in traditional communication systems.

In wireless communication network, handover could be classified into hard handover and soft handover. In long-term evolution (LTE) network, on the one hand, instead of radio network controller entity in the third generation (3G) network, eNodeB (eNB) performs handover [1]. Therefore the capability of centralised control disappears. On the other hand, LTE is an orthogonal frequency division multiplexing (OFDM) system and all the

cells share the same frequency band, thus user equipment (UE) cannot receive signals from adjacent base stations simultaneously as in the 3G system. As a result, LTE can only perform hard handover and the 'break before make' process must lead to an interruption and seriously affect the handover performance.

In cellular communication systems, in order to reduce the handover frequency and ping-pong effect, several optimised handover mechanisms have been proposed. A self-optimising handover algorithm based on pre-estimating the number of crossing cell-boundary is proposed in [2], which has a better performance in the rate of ping-pong handovers and the average number of handovers, and a simple handover forecasting scheme is introduced in [3], which minimises the number of handovers and lowers the ping-pong rate in LTE system. Meanwhile, some improvements to the assessment of handover performance in current handover schemes are made. The suitable value of reference signal received quality (RSRQ) during handover is evaluated in [4], and the linear L3 (Layer 3) filtering and carrier-to-interference ratio measurement is investigated in [5]; however, the improvement is limited just by parameter adjustment. A handover scheme in Manhattan scene is proposed in [6] which evaluates and compares the integrator algorithm with traditional power budget handover. A capacity integrated coordinated of multi-point (CoMP) handover algorithm supporting joint processing (JP) in LTE-advanced (LTE-A) system aiming to ensure the efficient use of radio resources and reduce unnecessary feedbacks is proposed in [7], and the effects

on performance and efficiency in LTE-A networks employing CoMP JP scheme and adaptive OFDM is proved in [8].

Owing to the particularity of the high-speed rail (HSR) communication, the current standardised handover scheme and optimised schemes in cellular network cannot reach a good handover performance. In recent years, some researches were carried out for improving the handover performance. A novel HSR access architecture to enhance handover performance is proposed in [9], which may represent the basic architecture of HSR research. A position and relay assisted robust handover scheme for HSR is put forward in [10], which improves the handover performance through enhancing the received signal strength within handover region, but can hardly change the nature of LTE hard handover, and there must be an interruption during 'break before make' process. On-vehicle dual-antenna handover scheme is introduced in [11, 12], which uses the dual-antenna to keep both connection to the serving eNB and the target eNB, but it is hard to implement and will need much modification to the train. The idea of soft handover for TD-LTE network in HSR scenario is proposed in [13], which needs to make a schedule in every transmission time interval, as a result, it lacks feasibility and increases the system complexity. A CoMP-based handover scheme for high-speed railway is presented in [14], which firstly introduces CoMP to handover. On the whole, the above work may, to some degree, improve the handover performance. However, it is found that most of the above work can hardly change the nature of LTE hard handover, but just enhance the received signal strength within handover region which optimises the handover triggering point and increases handover success rate. Thus, the enhancement is limited, and there must be an interruption during 'break before make' process. Besides, most of the above work only focuses on handover performance, which generally refers to handover success rate, without considering the handover delay, while handover delay is also an important indication during handover.

CoMP is an emerging recognised technology in LTE-A system [15]. The main principle of CoMP joint transmission (JT) is that the coordinated transmission points transmit the same signal through physical downlink shared channel or physical uplink shared channel adopting the same frequency and time resource to the receiver, and through pre-coding to convert the inter-cell interference into useful signal and thereby to achieve JT and diversity gain, which indicates that it can fundamentally solve the problem of LTE hard handover and satisfy the basic requirements of soft handover.

In LTE, as a candidate technology, bi-casting could be used to achieve the data sharing between serving eNB and target eNB, which means serving gateway (SGW) transmits the copied original data to both of serving eNB and target eNB through different paths. The interrupt time is short when bi-casting is used. However, if bi-casting is used directly into traditional LTE network, the problem of packet-loss will arise. At the beginning of bi-casting, there may be missing packets which are not successfully received by UE because of poor channel condition. Although UE accesses to the target eNB successfully, the target eNB transmits the data received from SGW directly, and the missed packets could never be received. Meanwhile, the additional signalling to execute bi-casting will increase the system complexity.

Compared with the current work mentioned above, it is found that if CoMP and bi-casting are combined, the handover delay can be reduced without packet-loss and complexity increase, and the advantage of CoMP-JT is taken to achieve soft handover. The remainder of this paper is organised as follows. System architecture and proposed handover scheme are introduced in Section 2. The simulation model of handover performance is presented and the handover delay is analysed in Section 3. Simulation results are given in Section 4. Section 5 concludes the whole paper.

2 System architecture and handover scheme

2.1 System architecture

As designed for HSR, the eNB is located along the rail in a linear mode. The system architecture is shown in Fig. 1, where the train is moving from eNB_i to eNB_j, eNB_i serves as the serving eNB and eNB_j serves as the target eNB.

To reduce penetration loss and avoid group handover, a mobile relay station (MRS) is installed on the roof of the train. As a result, the entire communication system is divided into two parts, namely on-vehicle part and ground-to-vehicle part. Inside the train, users communicate directly with access equipment such as femtocell base stations or wireless fidelity access points, and all the information is converged to MRS through on-board fibre. Ground-to-vehicle part is between MRS and eNB, which is the main concern in this paper.

2.2 Proposed CoMP and bi-casting joint handover scheme

Based on the system architecture mentioned above, CoMP is used on air interface and bi-casting is used on S1 interface. Different from traditional LTE handover process, the joining and exiting of CoMP and bi-casting processes are added before and after handover. In addition, the binding triggering of CoMP and bi-casting only needs one suit signalling, thus the system complexity is much simplified. The joining and exiting processes of CoMP and bi-casting are shown in Fig. 2, where UE stands for the MRS, and serving eNB (eNB_i) and target eNB (eNB_j) represent the two adjacent eNBs for implementing handover.

When the signal quality, reference signal received power (RSRP) or RSRQ, of the target eNB (eNB_j) is higher than that of serving eNB (eNB_i) a certain threshold, namely T_{add} , CoMP is triggered. The serving eNB initiates a 'cooperating request' message to the target eNB, which carries necessary information of UE and some cooperating related resource scheduling information. If the target eNB admits to join CoMP cooperating set, it reserves

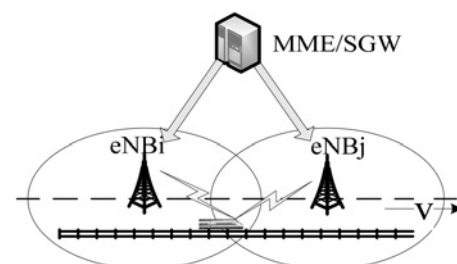


Fig. 1 System architecture

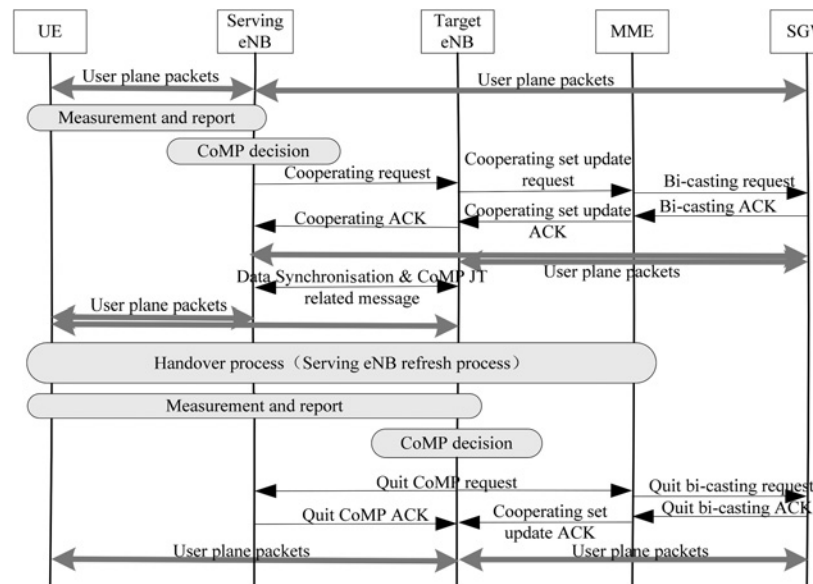


Fig. 2 Joining and exiting process of CoMP and bi-casting

corresponding resources and sends a 'cooperating set update' message to mobility management entity (MME) and notifies MME to trigger bi-casting to SGW. Although CoMP and bi-casting are activated, the SGW sends packets to both of the serving eNB and the target eNB simultaneously. The serving eNB and the target eNB buffer the packets, respectively, exchange necessary synchronisation data and CoMP-JT related information through X2 interface, and then transmit the buffered packets to UE simultaneously. After the handover process, when the signal quality of old serving eNB (eNB_i) is lower than that of the new serving eNB (eNB_j) a certain threshold, namely T_{quit} , the quit of CoMP is triggered. The new serving eNB sends 'quit CoMP request' message to the old serving eNB and 'cooperating set update request' message to MME. MME updates its communication context information and sends 'quit bi-casting request' message to SGW. SGW deletes the routing to the old serving eNB, and only delivers packets to the new serving eNB.

As much handover related information is stored in corresponding entity during the join process of CoMP and bi-casting, the proposed handover process is simplified and converted into update of serving eNB, which is illustrated in Fig. 3. As the triggering of CoMP is ahead of the triggering of handover, the value of T_{add} is smaller than handover threshold Γ . Although handover is triggered, the serving eNB sends 'radio resource control (RRC) connection reconfigure' message to UE and UE feeds back 'RRC connection reconfigure completed' message directly

to the target eNB since UE keeps connections to both eNBs in CoMP. Receiving 'RRC connection reconfigures completed' message, the target eNB changes its communication context and becomes a new serving eNB. Then the handover process is completed.

The main advantage of the proposed handover scheme is, firstly, based on the CoMP technology on air interface, the received signal quality is strengthened and a diversity gain is obtained. Moreover, the train keeps both connections to the serving eNB and the target eNB during handover, thus the soft handover is achieved. Secondly, based on the bi-casting used on S1 interface, SGW simultaneously sends data to the serving eNB and the target eNB without a data forwarding process on X2. Therefore the handover delay is greatly reduced.

3 Performance analysis

3.1 Handover performance analysis

In LTE, the measurement parameters include RSRP, received signal strength indicator and RSRQ. For simplicity, we choose RSRP for the handover decision. The strength of received signal is firstly analysed.

Suppose there are M sub-carriers in OFDM system, in one symbol duration, the transmitted signal can be given by

$$s_T(t) = \sum_{m=0}^{M-1} d_m(k) e^{j2\pi(f_0 + m\Delta f)t} \quad (1)$$

With channel fading and noise, the received signal strength can be denoted as

$$R_T(t) = \sqrt{\rho} s_T(t) \alpha(\tau, t) + n(t) \quad (2)$$

where $\alpha(\tau, t)$ is the total channel fading coefficient considering Doppler effect and multipath effect and $n(t)$ is the average noise power. $\rho = P_t \times \text{PL} \times \text{shadow}$ can be regarded as average received power after large-scale fading and P_t , PL and shadow represent the transmit power of eNB, path loss and shadow fading, respectively.

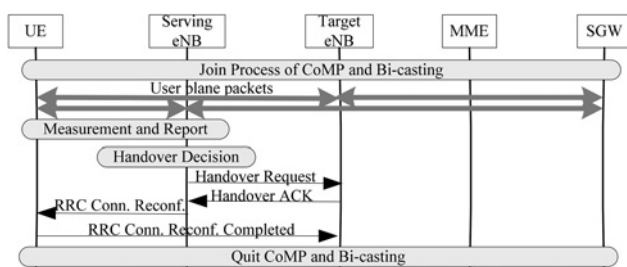


Fig. 3 Handover (update of serving eNB) process

When Doppler effect exists, the autocorrelation function of the n th path on m th sub-carrier is defined as

$$R_n(\tau) = \sigma_n^2 J_0(2\pi f_{d_n} \tau) \quad (3)$$

where $J_0(\cdot)$ is the zeroth-order Bessel function of first kind, f_{d_n} is the maximum Doppler frequency of the n th path and σ_n is the average receive power of small-scale fading of the n th path.

The output signal on the m th sub-carrier can be obtained as [12]

$$Y_m(k) = \frac{1}{T_s} \int_{kT_s}^{kT_s+T_s} R_T(t) \cdot e^{-j2\pi f_m(t-kT_s)} p(t-kT_s) dt \\ = \sqrt{\rho} \times \lambda_m^{\hat{m}}(k) d_m + \sum_{m=0, m \neq \hat{m}}^{M-1} \sqrt{\rho} \lambda_m^m(k) d_m + n(t) \quad (4)$$

where

$$\lambda_m^{\hat{m}}(k) = \frac{1}{T_s} \sum_{n=0}^{L-1} e^{-j2\pi(\hat{m}n/M)} \cdot \int_0^{T_s} \alpha'_n(t+kT_s) dt \quad (5)$$

$$E[|\lambda_m^{\hat{m}}(k)|^2] = \sum_{n=0}^{L-1} \sigma_n^2 \cdot \int_{-1}^1 J_0(2\pi f_{d_n} T_s x) (1-|x|) dx \quad (6)$$

where L is the number of resolvable multipath, T_s is the symbol period and $\alpha'_n(t+kT_s)$ is the channel fading coefficient of the n th path in one specified symbol period.

Thus, if the inter-carrier interference and noise are ignored, the strength of received signal is calculated as [12]

$$R = \rho \cdot E[|\lambda_m^{\hat{m}}(k)|^2] = P_t \times PL \times \text{shadow} \\ \times \sum_{n=0}^{L-1} \sigma_n^2 \cdot \int_{-1}^1 J_0(2\pi f_{d_n} T_s x) (1-|x|) dx \quad (7)$$

Based on the strength of received signal, the following handover performance can be obtained.

3.1.1 Handover outage probability: In LTE, time to trigger (TTT) is used to avoid ping-pong effect in handover

process, only the measurement result continuously meeting the handover trigger condition for a specific period of time, does the handover triggered.

When the TTT is not satisfied, the handover triggering probability is zero. Although satisfied, the handover triggering probability can be expressed as

$$P_x(i, j) = \Pr[10 \cdot \log_{10}(R_{(j,x)}) - 10 \cdot \log_{10}(R_{(i,x)}) \geq \Gamma] \quad (8)$$

where $R_{(j,x)}$ and $R_{(i,x)}$ represent the strength of received signals of the serving eNB and the target eNB, respectively, at the measurement point x . Although shadow = $10^{\varepsilon_{(j,x)}/10}$ and the shadow fading obeys Gaussian distribution, namely $\Pr[\varepsilon_{(j,x)} = \varepsilon_0] \sim N(0, \sigma_{(j,x)}^2)$, where $\sigma_{(j,x)}$ is the standard deviation of $\varepsilon_{(j,x)}$, (8) can be expressed as (see (9))

Outage probability refers to the probability of which receive signal quality being too weak to be properly decoded and leading to link interruption. Suppose when the received signal quality is lower than a certain threshold γ_{th} , the outage occurs.

In traditional handover scheme, the outage probability can be expressed as

$$P_x^{\text{outage}} = \Pr(\text{SINR}_x < \gamma_{th}) \quad (10)$$

where

$$\text{SINR}_x = \frac{R_{(i,x)}}{I_{(i,x)} + 10^{N_0 BW/10}} \quad (11)$$

and

$$I_{(i,x)} = \sum_{j=1, n \neq i}^{N_{BS}} R_{(j,x)} \quad (12)$$

Similar to handover triggering probability, according to the normal characteristics of shadow fading, the outage probability can be written as (see (13))

$$P_x(i, j) = \int_{-\infty}^{\infty} \Pr[10 \cdot \log_{10}(R_{(j,x)}) - 10 \cdot \log_{10}(R_{(i,x)}) \geq \Gamma | \varepsilon_{(j,x)} = \varepsilon_0] \cdot \Pr[\varepsilon_{(j,x)} = \varepsilon_0] d\varepsilon_0 \\ = \int_{-\infty}^{\infty} \Phi \left(\frac{10}{\sigma_{(i,x)}} \cdot \log \left(\frac{10^{[10 \cdot \log_{10}(R_{(j,x)}) - \Gamma]/10}}{P_t \cdot PL_{(i,BS,x)} \times \sum_{n=0}^{L-1} \sigma_n^2 \cdot \int_{-1}^1 J_0(2\pi f_{d_n} T_s x) (1-|x|) dx} \right) \right) \times \frac{1}{\sqrt{2\pi}\sigma_{(j,x)}} e^{-\varepsilon_0^2/(2\sigma_{(j,x)}^2)} d\varepsilon_0 \quad (9)$$

$$P_x^{\text{outage}} = \int_{-\infty}^{\infty} \Pr[(10 \cdot \log_{10} R_{(j,x)} - I_{(j,x)} - 10^{N_0 BW/10} \geq \gamma_{th}) | \varepsilon_{(j,x)} = \varepsilon_0] \cdot \Pr[\varepsilon_{(j,x)} = \varepsilon_0] d\varepsilon_0 \\ = \int_{-\infty}^{\infty} \Phi \left(\frac{\gamma_{th} + I_{(j,x)} + 10^{N_0 BW/10} - 10 \cdot \log_{10} \left(P_t \cdot PL_{(i,x)} \times \sum_{n=0}^{L-1} \sigma_n^2 \cdot \int_{-1}^1 J_0(2\pi f_{d_n} T_s x) (1-|x|) dx \right)}{\sigma_{(i,x)}} \right) \times \frac{1}{\sqrt{2\pi}\sigma_{(j,x)}} e^{-\varepsilon_0^2/(2\sigma_{(j,x)}^2)} d\varepsilon_0 \quad (13)$$

During the whole measurement process, the total outage probability at each measuring point is the addition of outage probability before and after handover, which can be expressed as

$$P_{\text{Total}}^{\text{outage}} = (1 - P_x(i, j)) \times \Pr(\text{SINR}_i < \gamma_{\text{th}}) + P_x(i, j) \times \Pr(\text{SINR}_j < \gamma_{\text{th}}) \quad (14)$$

In CoMP and bi-casting joint handover scheme proposed in this paper, the handover process is converted into update of serving eNB, both the serving eNB and the target eNB transmit the same packets to UE and UE can handle the received packets simultaneously during the whole process. As a result, only when both the received signal strength of serving eNB and the target eNB are lower than γ_{th} , the outage occurs, which can be expressed as

$$P_{\text{CoMP}}^{\text{outage}} = \Pr(\text{SINR}_{i \& j} < \gamma_{\text{th}}) \quad (15)$$

3.1.2 Handover success rate: The handover success rate refers to the probability that the received signal is properly decoded at each handover point, which is expressed as

$$P_x^{\text{success}} = P_x(i, j) \times (1 - P_x^{\text{outage}}) \quad (16)$$

3.1.3 System throughput: The system throughput is calculated in the Shannon formula, which is expressed as

$$\text{Throughput} = B \log_2(1 + \text{SINR}_x) \quad (17)$$

3.2 Handover delay analysis

The handover delay analysis is studied referred to [16, 17]. Suppose CoMP and bi-casting are triggered successfully, and the hybrid automatic repeat request (HARQ) repeat probability $p=0.3$ and the process number of repeat times $n=3$. Three handover schemes are compared: traditional LTE handover scheme [1], CoMP-based handover scheme [14] and CoMP and bi-casting joint scheme. The same handover preparation phase is omitted in comparison and the results are shown in Table 1.

Table 1 Handover delay comparison

Numbers	Signalling	Traditional LTE handover [1], ms	CoMP-based handover [14], ms	Proposed handover, ms
1	down link (DL) allocation	1	1	1
2	HARQ retransmission	7.2	<7.2	<7.2
3	RRC conn. reconf	1	1	1
4	HARQ retransmission	7.2	<7.2	<7.2
5	synchronisation	1	0	0
6	random access	20	0	0
7	eNB processing	4	0	0
8	up link (UL) allocation + timing advance (TA)	1	1	1
9	HARQ retransmission	7.2	<7.2	<7.2
10	UE processing	5	5	5
11	RRC conn. reconf. completed	1	1	1
12	sequence number (SN) status transmission	T_{X2-c}	T_{X2-c}	T_{X2-c}
13	data forwarding	T_{X2-U}	T_{X2-U}	0
14	target eNB processing	4	0	0
15	routing update	$2 \times T_{s1-c}$	$2 \times T_{s1-c}$	0
16	MME processing	15	15	0
17	routing update	$2 \times T_{s11}$	$2 \times T_{s11}$	0
18	SGW processing	15	15	0
19	eNB processing	4	0	0
20	UE context release	1	0	0
	total	$94.6 + 2 \times T_{s1-c} + 2 \times T_{s11}$	$<64.6 + 2 \times T_{s1-c} + 2 \times T_{s11}$	<30.6

Process numbers 11-12 and 5-10 are performed at the same time, so the delay should be the larger of them, and here suppose $X2$ has enough bandwidth and the delay of process numbers 11-12 is less than that in 5-10, and the total handover delay is calculated in Table 1. Compared with the traditional LTE scheme and CoMP-based handover scheme, more than 67% and 52.6% of the total time delay has been saved through the proposal analysed in this paper. Meanwhile, in user plane, there is about 30 ms interruption during process numbers 5-6 in traditional LTE handover, which does not exist in other two schemes. In addition, the processing delay between eNB and the core network, and the $X2$ data forwarding delay have been omitted.

4 Simulation and analysis

Based on the theoretical analysis, the proposed scheme is simulated and compared with traditional scheme. Detailed simulation parameters are shown in Table 2, referring to the actual scenario of HSR.

Using the scenario in Fig. 1, taking the vertical point along the track of the serving eNB as the starting point and the target eNB as the end point, the sampling points between them are measured.

Table 2 Simulation parameters

Parameters	Values
bandwidth	10 MHz
antenna pattern	Omni
carrier frequency	2.5 GHz
eNB transmit power	46 dBm
train antenna height	4 m
eNB antenna height	35 m
vertical distance between eNB and track	30 m
cell radius	3 km
site-to-site distance	4.8 km
shadow fading standard deviation	4 dB
path loss model	ITU-R M.1225
train speed	350 km/h
measurement period	200 ms
TTT	600 ms

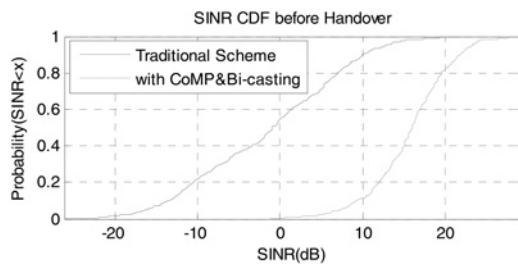


Fig. 4 SINR CDF before handover

The curve of signal to interference plus noise ratio (SINR) cumulative distribution function (CDF) before handover is shown in Fig. 4. From the figure, it can be seen that the SINR from the serving eNB in traditional scheme is in serious deterioration, the $\text{SINR} < 0$ part almost accounts for more than 50%, and it could be inferred that the closer to the handover point, the lower SINR is received by UE, which may lead to incorrect decoding. Although in the proposed scheme, since UE can simultaneously accept signals from both of serving eNB and target eNB, a diversity gain is achieved and the received SINR is much better.

The handover outage probability on each measurement point is shown in Fig. 5. In the traditional scheme, because of the weak signal received from the target eNB and the serving eNB in handover area, the outage probability is up to 50%. Although in the proposed scheme, the outage probability is decreased to $< 0.05\%$.

The cumulative distribution of handover success rate on each measurement point is shown in Fig. 6. The proposed scheme gains more than 20% improvement than that of traditional scheme. In Fig. 6, as TTT is added into simulation, the handover triggering probability is zero before handover, so the handover success rate is also zero.

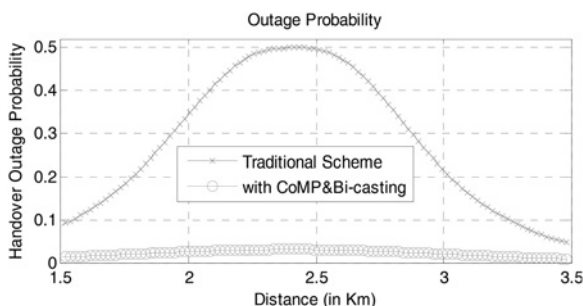


Fig. 5 Handover outage probability

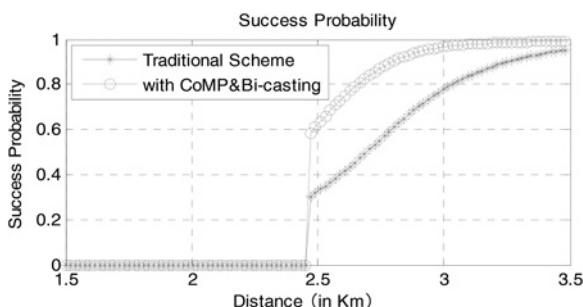


Fig. 6 Handover success rate

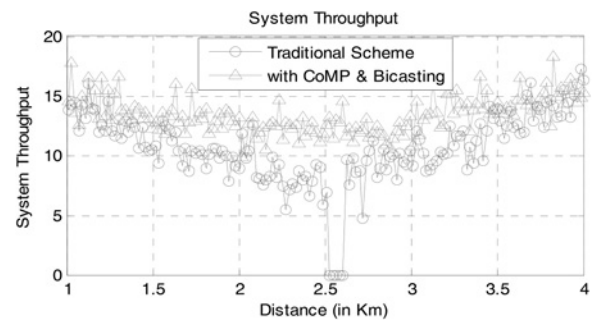


Fig. 7 System throughput

And when handover is triggered, the handover success rate increases gradually and ultimately reaches 1.

In Fig. 7, the system throughput is compared. The enhancement of the proposal is mainly reflected in handover region. In traditional scheme, UE only receives signal from either serving eNB or target eNB, both of which are under serious deterioration. Thus there is also a decrease in system throughput. Although in the proposal, the received signal strength is enhanced and the throughput has $\sim 30\%$ improvement. Meanwhile, as UE keeps being connected to both serving eNB and target eNB, the interruption during handover in user plane is effectively avoided.

5 Conclusion

With the expansion of the HSR industry and the improvement of railway speed worldwide, the reliability of ground-to-vehicle communication attracts the public attention. It is a great challenge to design an efficient handover scheme of broadband wireless communication with short interrupt time and high handover performance. In this paper, a soft handover scheme based on CoMP and bi-casting is proposed. It can be seen from theoretical analysis and simulation results, compared with traditional handover scheme, the handover delay is reduced more than 67%, the handover outage probability is decreased to $< 0.05\%$, the handover success rate is improved by about 20% and the system throughput in handover area is increased by 30% with zero interruption in the user plane, which strongly proves that the proposed CoMP and bi-casting joint handover scheme can guarantee the reliability of ground-to-vehicle communication as a much more feasible handover solution for HSR.

6 Acknowledgments

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7 References

- 3GPP TS 36.331 V10.2.0 (2011-06), Radio Resource Control (RRC) Protocol specification (Release 10)
- Zhang, H., Wen, X., Wang, B., Zheng, W., Lu, Z.: 'A novel self-optimizing handover mechanism for multi-service provisioning in

- LTE-advanced'. Int. Conf. Research Challenges in Computer Science, December 2009, pp. 221–224
- 3 Ge, H., Wen, X., Zheng, W., Lu, Z., Wang, B.: 'A history-based handover prediction for LTE systems'. Int. Symp. Computer Network and Multimedia Technology, December 2009, pp. 1–4
 - 4 Kurjenniemi, J., Henttonen, T., Kaikkonen, J.: 'Suitability of RSRQ measurement for quality based inter-frequency handover in LTE'. IEEE Int. Symp. Wireless Communication Systems, October 2008, pp. 703–707
 - 5 Anas, M., Calabrese, F.D., Ostling, P.E., Pedersen, K.I., Mogensen, E.: 'Performance analysis of handover measurements and layer-3 filtering for UTRAN LTE'. IEEE 18th Int. Symp. Personal, Indoor and Mobile Radio Communications, September 2007, pp. 1–5
 - 6 Zheng, N., Wigard, J.: 'On the performance of integrator handover algorithm in LTE networks'. IEEE 68th Vehicular Technology Conf. (VTC), September 2008, pp. 1–5
 - 7 Lin, C.-C., Sandrasegaran, K., Zhu, X., Xu, Z.: 'On the performance of capacity integrated CoMP handover algorithm in LTE-advanced'. 18th Asia-Pacific Conf. Communications (APCC 2012), October 2012, pp. 871–876
 - 8 Shrivastava, R., Aguayo-Torres, M.C.: 'Analysis and evaluation of cooperative multi-point transmission/reception and soft handover for LTE-advanced'. Information and Communication Technologies (WICT), Amity University, Noida, India, 30 October–2 November 2012, pp. 826–830
 - 9 Zhang, J.-Y., Tan, Z.-H., Zhong, Z.-D., Kong, Y.: 'A multi-mode multi-band and multi-system-based access architecture for high-speed railways'. IEEE 72nd Vehicular Technology Conf. Fall (VTC 2010-Fall), 2010, pp. 1–5
 - 10 Lu, L., Fang, X., Cheng, M., Yang, C., Luo, W., Di, C.: 'Positioning and relay assisted robust handover scheme for high speed railway'. IEEE 73rd Vehicular Technology Conf. (VTC Spring), 2011, pp. 1–5
 - 11 Yang, C., Lu, L., Di, C., Fang, X.: 'An on-vehicle dual-antenna handover scheme for high-speed railway distributed antenna system'. Int. Conf. Wireless Communications Networking and Mobile Computing (WiCOM), September 2010, pp. 1–5
 - 12 Tian, L., Zhou, Y., Li, J., Huang, Y., Shi, J., Zhou, J.: 'A novel handover scheme for seamless wireless connectivity in high-speed rail'. IEEE Seventh Int. Conf. Wireless and Mobile Computing, Networking and Communications (WiMob), 2011, pp. 230–236
 - 13 Wang, Q., Ren, G., Tu, J.: 'A soft handover algorithm for TD-LTE system in high-speed railway scenario'. IEEE Int. Conf. Signal Processing, Communications and Computing (ICSPCC), 2011, pp. 1–4
 - 14 Luo, W., Zhang, R., Fang, X.: 'A CoMP soft handover scheme for LTE systems in high speed railway', *EURASIP J. Wirel. Commun. Netw.*, 2012, **196**
 - 15 3GPP TR 36.819 V11.1.0 (2011–12), Coordinated multi-point operation for LTE physical layer aspects (Release 11)
 - 16 3GPP TR 25.912 V8.0.0 (2011–3), Feasibility study for evolved Universal Terrestrial Radio Access (UTRA) and Universal Terrestrial Radio Access Network (UTRAN) (release 10)
 - 17 3GPP TR 36.912 V10.0.0 (2011–03), Feasibility study for Further Advancements for E-UTRA (LTE-Advanced) (Release 10)